

Laboratory 2 - Diving deeper with Neo4j

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 - Version 1.0
-

- namespace: hertl-adv-daba-26
 - repo git: <https://github.com/nicoh1801/mse-advDaBa-Labo2>
 - ID exact du pod Neo4j: neo4j-6b6c5dc9f4-tw5qb
 - ID du pod/job loader: neo4j-loader-psfc7
 - Credentials Neo4j: User: neo4j Password: test
-

L'objectif global de ce travail pratique est de charger progressivement le dataset SBLP dans Neo4j.

Le dataset est fournis sous la forme d'un fichier JSONL. Chaque ligne correspond à un article. Le but est de transformer ces données en graphe Neo4j avec:

- des noeuds "article"
- des noeuds "author"
- des relations "authored" entre auteurs et articles
- des relations "cites" entre articles

Chargement des données avec le loader Java

Le chargement est fait par la classe [Loader.java](#). Cette classe lit les données depuis une URL, extrait les articles, auteurs et citations puis les insère dans Neo4j par lots.

Configuration par variables d'environnement

Le loader ne contient pas de config fixe. Les paramètres importants sont lus depuis les variables d'environnement.

```
String jsonUrl = getenv("JSON_FILE_URL", "http://vmrum.isc.heia-fr.ch/files/test.jsonl");
int maxArticles = Integer.parseInt(getenv("MAX_NODES", "10000"));
int batchSize = Integer.parseInt(getenv("BATCH_SIZE", "1000"));

String neo4jIp = getenv("NEO4J_IP", "localhost");
String neo4jUser = getenv("NEO4J_USER", "neo4j");
String neo4jPassword = getenv("NEO4J_PASSWORD", "test");
```

Configuration par variables d'environnement

Comme décrit dans l'Exemple.java qui a été fourni, le loader se connecte à Neo4j avec le driver officiel Neo4j java et le port 7687 qui est le port Bolt de Neo4j qui est utilisé par l'app Java pour envoyer des requêtes Cypher.

```
Driver driver = GraphDatabase.driver(
    "bolt://" + neo4jIp + ":7687",
    AuthTokens.basic(neo4jUser, neo4jPassword)
);
```

Création des contraintes

Avant d'insérer les données, le programme crée deux contraintes permettant d'éviter de créer plusieurs fois le même article ou le même auteur.

```
private static void createConstraints(Session session) {
    session.writeTransaction(tx -> {
        tx.run(
            "CREATE CONSTRAINT article_id IF NOT EXISTS " +
            "FOR (a:Article) " +
            "REQUIRE a._id IS UNIQUE"
        );
        tx.run(
            "CREATE CONSTRAINT author_id IF NOT EXISTS " +
            "FOR (a:Author) " +
            "REQUIRE a._id IS UNIQUE"
        );
        return null;
    });
}
```

Lecture du fichier JSONL en streaming

Le fichier est lu ligne par ligne depuis l'URL configurée. Chaque ligne est ensuite convertie en objet JSON.

```
URL url = new URL(jsonUrl);
BufferedReader br = new BufferedReader(new InputStreamReader(url.openStream()));

String line;

while ((line = br.readLine()) != null && articleCount < maxArticles) {
    JsonNode json = mapper.readTree(line);
}
```

Extraction des articles

Pour chaque ligne, le loader récupère l'identifiant et le titre de l'article. Si l'article possède un identifiant valide, il est ajouté au batch. L'identifiant JSON est utilisé comme priorité `_id` dans Neo4j.

```
String articleId = getText(json, "id");
String title = getText(json, "title");

if (articleId == null || articleId.trim().isEmpty()) {
    continue;
}

Map<String, Object> articleMap = new HashMap<String, Object>();
articleMap.put("id", articleId);
articleMap.put("title", title == null ? "" : title);
articleBatch.add(articleMap);
```

Extraction des auteurs

Les auteurs sont lus depuis le champ `authors`. Pour chaque auteur, le programme récupère son identifiant et son nom.

```
JsonNode authors = json.get("authors");
if (authors != null && authors.isArray()) {
    for (JsonNode author : authors) {
        String authorId = getText(author, "id");
        String authorName = getText(author, "name");

        if (authorId == null || authorId.trim().isEmpty()) {
            continue;
        }

        uniqueAuthors.add(authorId);

        Map<String, Object> authorMap = new HashMap<String, Object>();
        authorMap.put("id", authorId);
        authorMap.put("name", authorName == null ? "" : authorName);
        authorBatch.add(authorMap);

        Map<String, Object> authoredMap = new HashMap<String, Object>();
        authoredMap.put("authorId", authorId);
        authoredMap.put("articleId", articleId);
        authoredBatch.add(authoredMap);
    }
}
```

Extraction des citations

Les citations quant à elle sont lues depuis le champ `references`. Pour chaque citation le loader prépare une relation entre l'article courant et l'article cité. Même si l'article cité n'a pas encore été lu dans le fichier, il peut

déjà être créé dans Neo4j avec son `_id`. Son titre sera complété plus tard si l'article apparaît dans le dataset.

```
JsonNode references = json.get("references");
if (references != null && references.isArray()) {
    for (JsonNode ref : references) {
        if (ref == null || !ref.isTextual()) {
            continue;
        }

        String citedId = ref.asText();

        Map<String, Object> citationMap = new HashMap<String, Object>();
        citationMap.put("articleId", articleId);
        citationMap.put("citedId", citedId);
        citationBatch.add(citationMap);
    }
}
```

insertion par batchs

Pour éviter d'envoyer une requête par ligne, les données sont insérées par batch. la taille du batch est configurable avec `BATCH_SIZE`.

```
if (articleCount % batchSize == 0) {
    insertBatch(session, articleBatch, authorBatch, authoredBatch, citationBatch);

    long elapsed = (System.currentTimeMillis() - start) / 1000;

    System.out.println(
        "PROGRESS articles=" + articleCount +
        " authors=" + uniqueAuthors.size() +
        " totalNodes=" + (articleCount + uniqueAuthors.size()) +
        " elapsedSeconds=" + elapsed
    );

    articleBatch.clear();
    authorBatch.clear();
    authoredBatch.clear();
    citationBatch.clear();
}
```

à chaque batch, les articles, auteurs, relation authored et cites sont insérés. UNWIND permet d'envoyer une liste de données à Neo4j et de les traiter côté db.

```
private static void insertBatch(
    Session session,
    final List<Map<String, Object>> articles,
    final List<Map<String, Object>> authors,
```

```

        final List<Map<String, Object>> authored,
        final List<Map<String, Object>> citations
    ) {
        session.writeTransaction(tx -> {
            tx.run(
                "UNWIND $articles AS row " +
                "MERGE (a:Article {_id: row.id}) " +
                "SET a.title = row.title",
                parameters("articles", articles)
            );

            tx.run(
                "UNWIND $authors AS row " +
                "MERGE (a:Author {_id: row.id}) " +
                "SET a.name = row.name",
                parameters("authors", authors)
            );

            tx.run(
                "UNWIND $authored AS row " +
                "MATCH (author:Author {_id: row.authorId}) " +
                "MATCH (article:Article {_id: row.articleId}) " +
                "MERGE (author)-[:AUTHORED]->(article)",
                parameters("authored", authored)
            );

            tx.run(
                "UNWIND $citations AS row " +
                "MATCH (article:Article {_id: row.articleId}) " +
                "MERGE (cited:Article {_id: row.citedId}) " +
                "MERGE (article)-[:CITES]->(cited)",
                parameters("citations", citations)
            );

            return null;
        });
    }

```

Déploiement local avec Docker Compose

Pour tester localement, un docker compose est utiliser avec deux conteneurs, un db contenant Neo4j et un app contenant le loader java. Celui-ci a en grande partie été mis à disposition dans le repo de base.

```

services:
  db:
    image: neo4j:4.4.15-community
    ports:
      - "7474:7474"
      - "7687:7687"
    networks:
      internal:

```

```

    ipv4_address: 172.24.0.10
  volumes:
    - $PWD/neo4j_mount/data:/data
    - $PWD/neo4j_mount/logs:/logs
    - $PWD/neo4j_mount/conf:/conf
  environment:
    - NEO4J_AUTH=neo4j/test
  deploy:
    resources:
      limits:
        memory: 3g

app:
  image: neo4jtp:latest
  depends_on:
    - db
  networks:
    - internal
  volumes:
    - $PWD/dblpExample.jsonl:/file.jsonl
  environment:
    - JSON_FILE_URL=http://vmrum.isc.heia-fr.ch/files/DBLP-Citation-network-
V18.jsonl
    - MAX_NODES=10000
    - NEO4J_IP=172.24.0.10 # must be the same as above
    - NEO4J_USER=neo4j
    - NEO4J_PASSWORD=test
    - BATCH_SIZE=1
  deploy:
    resources:
      limits:
        memory: 4g

networks:
  internal:
    ipam:
      driver: default
      config:
        - subnet: "172.24.0.0/24"

```

Premier résultat local

Avec la configuration suivante:

```

- JSON_FILE_URL=http://vmrum.isc.heia-fr.ch/files/DBLP-Citation-network-V18.jsonl
- MAX_NODES=10000
- BATCH_SIZE=1000

```

Avant le test, la base local Neo4j a été supprimée afin de repartir à vide:

```
docker compose down
Remove-Item -Recurse -Force .\neo4j_mount
docker compose up
```

Le loader a ensuite lu le fichier JSONL depuis l'url et a inséré les data dans neo4j par batch de 1000 acticles. Les logs sont les suivants:

```
LOAD_START=2026-05-02T17:27:36.256127980Z
JSON_URL=http://vmrum.isc.heia-fr.ch/files/DBLP-Citation-network-V18.jsonl
MAX_ARTICLES=10000
BATCH_SIZE=1000
NEO4J_IP=172.24.0.10

PROGRESS articles=1000 authors=2584 totalNodes=3584 elapsedSeconds=46
PROGRESS articles=2000 authors=5114 totalNodes=7114 elapsedSeconds=48
PROGRESS articles=3000 authors=7934 totalNodes=10934 elapsedSeconds=50
PROGRESS articles=4000 authors=11001 totalNodes=15001 elapsedSeconds=52
PROGRESS articles=5000 authors=14351 totalNodes=19351 elapsedSeconds=53
PROGRESS articles=6000 authors=17803 totalNodes=23803 elapsedSeconds=55
PROGRESS articles=7000 authors=20858 totalNodes=27858 elapsedSeconds=56
PROGRESS articles=8000 authors=23265 totalNodes=31265 elapsedSeconds=57
PROGRESS articles=9000 authors=25713 totalNodes=34713 elapsedSeconds=57
PROGRESS articles=10000 authors=28309 totalNodes=38309 elapsedSeconds=58

LOAD_END=2026-05-02T17:28:34.826419941Z
ARTICLES_LOADED=10000
AUTHORS_LOADED=28309
TOTAL_NODES=38309
DURATION_SECONDS=58
```

Pour ce test local, le program à donc chargé:

```
Articles : 10 000
Auteurs distincts : 28 309
Total de noeuds : 38 309
Durée : 58 secondes
```

Quelques vérifications directement via l'interface de Neo4j ont été fait afin de voir le bon chargement des datas.

The screenshot shows the Neo4j Browser interface. The address bar indicates the connection to `neo4j@neo4j://localhost:7687/`. The browser is displaying a Cypher query: `neo4j$ MATCH (a:Article) RETURN a._id, a.title LIMIT 10;`. The results are shown in a table with two columns: `a._id` and `a.title`. The table contains 10 rows of data. A status message at the bottom indicates: "Started streaming 10 records after 8 ms and completed after 12 ms."

	a._id	a.title
1	"5390877920f70186a0d2ce7f"	"Top-Down Construction of 3-D Mechanical Object Shapes from Engineering Drawings"
2	"5390877920f70186a0d2d85b"	"MH: A Multifarious User Agent."
3	"5390877920f70186a0d2e433"	"Characterizing an Optimal Input in Perturbed Convex Programming: Corrigendum"
4	"5390877f20f70186a0d2fca6"	"Theoretical Aspects of Sigma-programming"
5	"5390878320f70186a0d33e0e"	"I 3 R: a New Approach to the Design of Document Retrieval Systems"
6	"5390878720f70186a0d342f2"	"The Use of Fuzzy Information Retrieval Techniques in Construction of Multi-Centre Knowledge-Based Systems."
7		

The screenshot shows the Neo4j Browser interface. The address bar indicates the connection to `neo4j@neo4j://localhost:7687/`. The browser is displaying a Cypher query: `neo4j$ MATCH ()-[r]->() RETURN type(r) AS relationshipType, count(r) AS count;`. The results are shown in a table with two columns: `relationshipType` and `count`. The table contains 2 rows of data. A status message at the bottom indicates: "Started streaming 2 records after 7 ms and completed after 269 ms."

	relationshipType	count
1	"CITES"	98745
2	"AUTHORED"	29584

Déploiement sur Kubernetes

Le déploiement final a été réalisé sur le cluster Kubernetes de l'école, dans le namespace **hertl-adv-daba-26**. L'objectif du déploiement est d'avoir deux composants séparés. Un pod Neo4j qui contient la base de

données et conserve les données chargées et un job kubernetes neo4j-loader qui lance le programme java pour le chargement.

Contrairement au déloiement local avec docker compose, le loader ne se connecte pas à neo4j avec une IP fixe. Dans kubernetes, je créer un service nommé neo4j qui permettra au pod du loader de résoudre automatiquement l'adresse du pod Neo4j via le dns interne de kubernetes.

```
- name: NEO4J_IP  
  value: "neo4j"
```

Image docker du loader

Pour que Kubernetes puisse lancer le loader, l'image Docker ne peut pas rester uniquement sur la machine local. Elle doit être disponible dans un registry docker. l'image local a donc été taguée puis poussée sur Docker Hub:

```
docker tag neo4jtp:latest nicolasdockerhertling/neo4jtp:latest  
docker push nicolasdockerhertling/neo4jtp:latest
```

L'image est utilisée après dans le job kubernetes:

```
image: nicolasdockerhertling/neo4jtp:latest
```

Fichiers Kubernetes

Les fichiers Kubernetes ont été placés dans le dossier [k8s/](#).

```
k8s/neo4j-pvc.yaml  
k8s/neo4j-deployment.yaml  
k8s/neo4j-service.yaml  
k8s/loader-job.yaml
```

Le fichier [neo4j-pvc.yaml](#) créer un volume persistant pur conserver les data Neo4j.

Le fichier [neo4j-deployment.yaml](#) lance le pod Neo4j avec l'image officiel.

[neo4j-service.yaml](#) expose Neo4j à l'intérieur du namespace Kubernetes notamment sur le port bolt 7687.

Finalement, le fichier [loader-job.yaml](#) lance le programme java de chargement. Les données sont chargées via l'url "http://vmrum.isc.heia-fr.ch/files/DBLP-Citation-network-V18.jsonl", avec les paramètres suivants:

```
- name: JSON_FILE_URL  
  value: "http://vmrum.isc.heia-fr.ch/files/DBLP-Citation-network-V18.jsonl"
```

```
- name: MAX_NODES
  value: "1000000"
- name: BATCH_SIZE
  value: "1000"
- name: NEO4J_IP
  value: "neo4j"
- name: NEO4J_USER
  value: "neo4j"
- name: NEO4J_PASSWORD
  value: "test"
```

Déploiement de Neo4j

Les ressources Neo4j sont créées avec les commande suivantes:

```
kubectl apply -f k8s/neo4j-pvc.yaml -n hertl-adv-daba-26
kubectl apply -f k8s/neo4j-deployment.yaml -n hertl-adv-daba-26
kubectl apply -f k8s/neo4j-service.yaml -n hertl-adv-daba-26
```

Après le déploiement, on peut vérifier relativement facilement si le bod Neo4j a bien démarré:

```
kubectl get pods -n hertl-adv-daba-26
```

NAME	READY	STATUS	RESTARTS	AGE
neo4j-6b6c5dc9f4-92dk7	1/1	Running	0	13s

Déploiement du Loader

Le loder est déployé sous fome de Job. Ce qui signifie qu'il s'exécute une fois, charge les données et se termine.

Pour lancer le job:

```
kubectl apply -f k8s/loader-job.yaml -n hertl-adv-daba-26
```

Voir logs:

```
kubectl logs job/neo4j-loader -n hertl-adv-daba-26
```

suvire logs:

```
kubectl logs -f job/neo4j-loader -n hertl-adv-daba-26
```

Redémarrage d'un test

Quand on modifie le loader-job.yaml, Kubernetes ne met pas automatiquement à jour un job existant. Il faut supprimer l'ancien job et le recréer:

```
kubectl delete job neo4j-loader -n hertl-adv-daba-26  
kubectl apply -f k8s/loader-job.yaml -n hertl-adv-daba-26
```

Pour remettre d'une base complètement vide, il faut aussi supprimer le déploiement neo4j et son volume persistant:

```
kubectl delete job neo4j-loader -n hertl-adv-daba-26  
kubectl delete deployment neo4j -n hertl-adv-daba-26  
kubectl delete pvc neo4j-data -n hertl-adv-daba-26
```

Et recréer:

```
kubectl apply -f k8s/neo4j-pvc.yaml -n hertl-adv-daba-26  
kubectl apply -f k8s/neo4j-deployment.yaml -n hertl-adv-daba-26  
kubectl apply -f k8s/neo4j-service.yaml -n hertl-adv-daba-26
```

et relancer le loader:

```
kubectl apply -f k8s/loader-job.yaml -n hertl-adv-daba-26
```

Streaming et batchs

Dans la consigne, on demande que les données soient streamées depuis l'URL fournie. Ce qui ne signifie pas qu'il faut insérer un article à la fois dans Neo4j. Afin de permettre les deux, le fichier JSONL est lu en streaming, ligne par ligne :

```

URL url = new URL(jsonUrl);
BufferedReader br = new BufferedReader(new InputStreamReader(url.openStream()));

String line;

while ((line = br.readLine()) != null && articleCount < maxArticles) {
    JsonNode json = mapper.readTree(line);

```

Le fichier complet n'est donc jamais copié dans le conteneur et n'est jamais chargé entièrement en mémoire.

Le paramètre **BATCH_SIZE** sert uniquement à grouper les insertions dans Neo4j. Avec un BATCH_SIZE de 1000, le programme lit tjrs les données en streaming, mais insère les données dans Neo4j tous les 1000 articles, permettant de diminuer fortement le nbre de transactions envoyées à la base et améliorant les performances.

Un essai avec BATCH_SIZE=1 a montré que cette configuration est beaucoup plus lente, car elle crée une transaction par article. Pour les tests significatifs, j'ai choisi d'utiliser BATCH_SIZE=1000.

Résultat Kubernetes avec 10 000 articles

un premier test kubernetes a été réalisé avec la configuration suivante:

```

- name: MAX_NODES
  value: "10000"
- name: BATCH_SIZE
  value: "1000"

```

Les logs importants sont les suivants:

```

LOAD_START=2026-05-03T19:39:25.877196755Z
JSON_URL=http://vmrum.isc.heia-fr.ch/files/DBLP-Citation-network-V18.jsonl
MAX_ARTICLES=10000
BATCH_SIZE=1000
NEO4J_IP=neo4j

PROGRESS articles=1000 authors=2584 totalNodes=3584 elapsedSeconds=8
PROGRESS articles=2000 authors=5114 totalNodes=7114 elapsedSeconds=9
PROGRESS articles=3000 authors=7934 totalNodes=10934 elapsedSeconds=11
PROGRESS articles=4000 authors=11001 totalNodes=15001 elapsedSeconds=13
PROGRESS articles=5000 authors=14351 totalNodes=19351 elapsedSeconds=14
PROGRESS articles=6000 authors=17803 totalNodes=23803 elapsedSeconds=16
PROGRESS articles=7000 authors=20858 totalNodes=27858 elapsedSeconds=18
PROGRESS articles=8000 authors=23265 totalNodes=31265 elapsedSeconds=19
PROGRESS articles=9000 authors=25713 totalNodes=34713 elapsedSeconds=20
PROGRESS articles=10000 authors=28309 totalNodes=38309 elapsedSeconds=21

LOAD_END=2026-05-03T19:39:47.659166507Z
ARTICLES_LOADED=10000

```

```
AUTHORS_LOADED=28309
TOTAL_NODES=38309
DURATION_SECONDS=21
```

Ce test a donc chargé:

- Articles complets lus : 10 000
- Auteurs distincts : 28 309
- Total articles + auteurs : 38 309
- Durée : 21 secondes
- Débit observé : $38\,309 / 21 =$ environ 1825 noeuds par secondes

Résultat Kubernetes avec 100 000 articles

Un second test a été réalisé avec la config suivante:

```
- name: MAX_NODES
  value: "100000"
- name: BATCH_SIZE
  value: "1000"
```

Les logs importants sont les suivants:

```
PROGRESS articles=100000 authors=235438 totalNodes=335438 elapsedSeconds=134
LOAD_END=2026-05-03T20:20:48.595254376Z
ARTICLES_LOADED=100000
AUTHORS_LOADED=235438
TOTAL_NODES=335438
DURATION_SECONDS=134
```

Ce test a donc chargé:

- Articles complets lus : 100 000
- Auteurs distincts : 235 438
- Total articles + auteurs : 335 438
- Durée : 134 secondes
- Débit observé : $335\,438 / 134 =$ environ 2503 noeuds par secondes

Résultat Kubernetes avec 1 000 000 articles

Un troisième test a été réalisé avec la config suivante:

```
- name: MAX_NODES
  value: "1000000"
```

```
- name: BATCH_SIZE
  value: "1000"
```

Les logs importants sont les suivants:

```
LOAD_START=2026-05-03T20:54:16.074691587Z
JSON_URL=http://vmrum.isc.heia-fr.ch/files/DBLP-Citation-network-V18.jsonl
MAX_ARTICLES=1000000
BATCH_SIZE=1000
NEO4J_IP=neo4j

LOAD_END=2026-05-03T21:59:26.386887899Z
ARTICLES_LOADED=1000000
AUTHORS_LOADED=1242546
TOTAL_NODES=2242546
DURATION_SECONDS=3910
```

Ce test a donc chargé:

- Articles complets lus : 1 000 000
- Auteurs distincts : 1 242 546
- Total articles + auteurs : 2 242 546
- Durée : 3 910 secondes, soit environ 65 minutes
- Débit observé : 2 242 546 / 3 910 = environ 573 noeuds par secondes

On peut voir que le débit est plus faible que pour le test à 100 000 articles, probablement dû au fait que la base devient beaucoup plus grande, ce qui rend les opérations MERGE plus coûteuses, notamment sur les articles cités et les relations.

Vérification des données dans Neo4j

Une vérification des données a été faite directement dans le pod neo4j avec cypher-shell:

```
kubectl get pods -n hertl-adv-daba-26
NAME                                READY   STATUS    RESTARTS   AGE
neo4j-6b6c5dc9f4-tw5qb             1/1     Running   0           88m
neo4j-loader-psfc7                  0/1     Completed 0           87m

kubectl exec -it neo4j-6b6c5dc9f4-tw5qb -n hertl-adv-daba-26 -- cypher-shell -u
neo4j -p test
```

Nombre de noeuds par label:

```
MATCH (n) RETURN labels(n), count(n);
+-----+
| labels(n) | count(n) |
+-----+
| ["Article"] | 3108245 |
| ["Author"] | 1242546 |
+-----+
```

Il y a plus d'articles est supérieur au nombre d'articles complets lus depuis le fichiers JSONL. ce qui est normal car les articles cités sont également créés lors de l'insertion des relation CITES, même si leur ligne complète n'a pas encore été lue.

Nombre de relation par type:

```
MATCH ()-[r]->() RETURN type(r), count(r);
+-----+
| type(r) | count(r) |
+-----+
| "CITES" | 10945109 |
| "AUTHORED" | 3077492 |
+-----+
```

Vérification de quelques articles:

```
MATCH (a:Article) RETURN a._id, a.title LIMIT 10;

| "5390877920f70186a0d2ce7f" | "Top-Down Construction of 3-D Mechanical Object Shapes from Engineering Drawings" |
| "5390877920f70186a0d2d85b" | "MH: A Multifarious User Agent." |
| "5390877920f70186a0d2e433" | "Characterizing an Optimal Input in Perturbed Convex Programming: Corrigendum" |
```

Quelques auteurs:

```
MATCH (a:Author) RETURN a._id, a.name LIMIT 10;

| "62aad3b2d9f2040d085dc9f9" | "Hiroshi Yoshiura" |
| "5608ec4a45cedb3396db2920" | "Kikuo Fujimura" |
| "548a62d3dabfae8a11fb49e2" | "Tosiyasu L. Kunii" |
```

En raison du temps de traitement, je n'ai pas testé avec plus de 10 millions d'articles, mais le principe reste le même.

Annexe: preuve des logs

```

PS C:\Users\LocalAdmin\MSE-
Local\TSM_AdvDaBa\TP02_Large_Database_Experiment_with_Neo4j\mse-advDaBa-Labo2>
kubectl logs job/neo4j-loader -n hertl-adv-daba-26
[INFO] Scanning for projects...
[WARNING]
[WARNING] Some problems were encountered while building the effective model for
icosys:advDB:jar:1.0-SNAPSHOT
[WARNING] 'dependencies.dependency.version' for org.junit.jupiter:junit-
jupiter:jar is either LATEST or RELEASE (both of them are being deprecated) @ line
38, column 22
[WARNING] 'build.plugins.plugin.version' for org.codehaus.mojo:exec-maven-plugin
is missing. @ line 93, column 21
[WARNING]
[WARNING] It is highly recommended to fix these problems because they threaten the
stability of your build.
[WARNING]
[WARNING] For this reason, future Maven versions might no longer support building
such malformed projects.
[WARNING]
Downloading from central:
https://repo.maven.apache.org/maven2/org/codehaus/mojo/exec-maven-plugin/maven-
metadata.xml
Downloaded from central:
https://repo.maven.apache.org/maven2/org/codehaus/mojo/exec-maven-plugin/maven-
metadata.xml (1.1 kB at 3.4 kB/s)
Downloading from central:
https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml
Downloading from central:
https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml
Downloaded from central:
https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml
(14 kB at 549 kB/s)
Downloaded from central:
https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml (21 kB
at 290 kB/s)
[INFO]
[INFO] -----< icosys:advDB >-----
[INFO] Building advDB 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] -----[ jar ]-----
Downloading from central:
https://repo.maven.apache.org/maven2/org/junit/jupiter/junit-jupiter/maven-
metadata.xml
Downloaded from central:
https://repo.maven.apache.org/maven2/org/junit/jupiter/junit-jupiter/maven-
metadata.xml (3.0 kB at 95 kB/s)
Downloading from central:
https://repo.maven.apache.org/maven2/org/junit/jupiter/junit-jupiter/6.1.0-
RC1/junit-jupiter-6.1.0-RC1.jar
Downloaded from central:

```



```
https://repo.maven.apache.org/maven2/org/junit/jupiter/junit-jupiter/6.1.0-RC1/junit-jupiter-6.1.0-RC1.jar (6.4 kB at 237 kB/s)
Downloading from central:
https://repo.maven.apache.org/maven2/org/junit/jupiter/junit-jupiter-api/6.1.0-RC1/junit-jupiter-api-6.1.0-RC1.jar
Downloading from central:
https://repo.maven.apache.org/maven2/org/opentest4j/opentest4j/1.3.0/opentest4j-1.3.0.jar
Downloading from central:
https://repo.maven.apache.org/maven2/org/junit/platform/junit-platform-commons/6.1.0-RC1/junit-platform-commons-6.1.0-RC1.jar
Downloading from central:
https://repo.maven.apache.org/maven2/org/apiguardian/apiguardian-api/1.1.2/apiguardian-api-1.1.2.jar
Downloading from central:
https://repo.maven.apache.org/maven2/org/jspecify/jspecify/1.0.0/jspecify-1.0.0.jar
Downloaded from central:
https://repo.maven.apache.org/maven2/org/jspecify/jspecify/1.0.0/jspecify-1.0.0.jar (3.8 kB at 73 kB/s)
Downloading from central:
https://repo.maven.apache.org/maven2/org/junit/jupiter/junit-jupiter-params/6.1.0-RC1/junit-jupiter-params-6.1.0-RC1.jar
Downloaded from central:
https://repo.maven.apache.org/maven2/org/apiguardian/apiguardian-api/1.1.2/apiguardian-api-1.1.2.jar (6.8 kB at 117 kB/s)
Downloading from central:
https://repo.maven.apache.org/maven2/org/junit/jupiter/junit-jupiter-engine/6.1.0-RC1/junit-jupiter-engine-6.1.0-RC1.jar
Downloaded from central:
https://repo.maven.apache.org/maven2/org/opentest4j/opentest4j/1.3.0/opentest4j-1.3.0.jar (14 kB at 234 kB/s)
Downloading from central:
https://repo.maven.apache.org/maven2/org/junit/platform/junit-platform-engine/6.1.0-RC1/junit-platform-engine-6.1.0-RC1.jar
Downloaded from central:
https://repo.maven.apache.org/maven2/org/junit/platform/junit-platform-commons/6.1.0-RC1/junit-platform-commons-6.1.0-RC1.jar (176 kB at 410 kB/s)
Downloaded from central:
https://repo.maven.apache.org/maven2/org/junit/jupiter/junit-jupiter-api/6.1.0-RC1/junit-jupiter-api-6.1.0-RC1.jar (312 kB at 706 kB/s)
Downloaded from central:
https://repo.maven.apache.org/maven2/org/junit/platform/junit-platform-engine/6.1.0-RC1/junit-platform-engine-6.1.0-RC1.jar (324 kB at 639 kB/s)
Downloaded from central:
https://repo.maven.apache.org/maven2/org/junit/jupiter/junit-jupiter-engine/6.1.0-RC1/junit-jupiter-engine-6.1.0-RC1.jar (353 kB at 676 kB/s)
Downloaded from central:
https://repo.maven.apache.org/maven2/org/junit/jupiter/junit-jupiter-params/6.1.0-RC1/junit-jupiter-params-6.1.0-RC1.jar (303 kB at 554 kB/s)
[INFO]
[INFO] --- exec:3.6.3:java (default-cli) @ advDB ---
Downloading from central:
https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-
```

```
util/1.9.24/maven-resolver-util-1.9.24.pom
Downloaded from central:
https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-
util/1.9.24/maven-resolver-util-1.9.24.pom (2.2 kB at 101 kB/s)
Downloading from central:
https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-
resolver/1.9.24/maven-resolver-1.9.24.pom
Downloaded from central:
https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-
resolver/1.9.24/maven-resolver-1.9.24.pom (25 kB at 1.3 MB/s)
Downloading from central:
https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/45/maven-
parent-45.pom
Downloaded from central:
https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/45/maven-
parent-45.pom (53 kB at 2.3 MB/s)
Downloading from central:
https://repo.maven.apache.org/maven2/org/apache/apache/35/apache-35.pom
Downloaded from central:
https://repo.maven.apache.org/maven2/org/apache/apache/35/apache-35.pom (24 kB at
1.2 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/junit/junit-
bom/5.13.1/junit-bom-5.13.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/junit/junit-
bom/5.13.1/junit-bom-5.13.1.pom (5.6 kB at 332 kB/s)
Downloading from central:
https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-
api/1.9.24/maven-resolver-api-1.9.24.pom
Downloaded from central:
https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-
api/1.9.24/maven-resolver-api-1.9.24.pom (2.2 kB at 117 kB/s)
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utils/4.0.2/plexus-utils-4.0.2.pom
Downloaded from central:
https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-
utils/4.0.2/plexus-utils-4.0.2.pom (13 kB at 557 kB/s)
Downloading from central:
https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-xml/3.0.1/plexus-
xml-3.0.1.pom
Downloaded from central:
https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-xml/3.0.1/plexus-
xml-3.0.1.pom (3.7 kB at 205 kB/s)
Downloading from central:
https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus/18/plexus-18.pom
Downloaded from central:
https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus/18/plexus-18.pom
(29 kB at 1.4 MB/s)
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https://repo.maven.apache.org/maven2/org/apache/commons/commons-
exec/1.6.0/commons-exec-1.6.0.pom
Downloaded from central:
https://repo.maven.apache.org/maven2/org/apache/commons/commons-
exec/1.6.0/commons-exec-1.6.0.pom (11 kB at 555 kB/s)
```

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Downloaded from central: <https://repo.maven.apache.org/maven2/org/junit/junit-bom/5.14.1/junit-bom-5.14.1.pom> (5.7 kB at 283 kB/s)
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<https://repo.maven.apache.org/maven2/org/ow2/asm/asm/9.9.1/asm-9.9.1.pom>
Downloaded from central:
<https://repo.maven.apache.org/maven2/org/ow2/asm/asm/9.9.1/asm-9.9.1.pom> (2.4 kB at 108 kB/s)
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Downloaded from central:
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Downloading from central: <https://repo.maven.apache.org/maven2/org/ow2/asm/asm-commons/9.9.1/asm-commons-9.9.1.pom>
Downloaded from central: <https://repo.maven.apache.org/maven2/org/ow2/asm/asm-commons/9.9.1/asm-commons-9.9.1.pom> (2.8 kB at 155 kB/s)
Downloading from central: <https://repo.maven.apache.org/maven2/org/ow2/asm/asm-tree/9.9.1/asm-tree-9.9.1.pom>
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<https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.24/maven-resolver-util-1.9.24.jar>
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<https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.24/maven-resolver-util-1.9.24.jar> (196 kB at 4.4 MB/s)
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Downloading from central:
<https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/4.0.2/plexus-utils-4.0.2.jar>
Downloading from central:
<https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-xml/3.0.1/plexus-xml-3.0.1.jar>
Downloading from central:
<https://repo.maven.apache.org/maven2/org/apache/commons/commons-exec/1.6.0/commons-exec-1.6.0.jar>
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https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-
api/1.9.24/maven-resolver-api-1.9.24.jar (157 kB at 4.8 MB/s)
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tree/9.9.1/asm-tree-9.9.1.jar
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utils/4.0.2/plexus-utils-4.0.2.jar (193 kB at 5.3 MB/s)
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at 2.9 MB/s)
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xml-3.0.1.jar (94 kB at 1.8 MB/s)
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tree/9.9.1/asm-tree-9.9.1.jar (52 kB at 1.0 MB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm-
commons/9.9.1/asm-commons-9.9.1.jar (75 kB at 1.4 MB/s)
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JSON_URL=http://vmrum.isc.heia-fr.ch/files/DBLP-Citation-network-V18.jsonl
MAX_ARTICLES=1000000
BATCH_SIZE=1000
NEO4J_IP=neo4j
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::objectFieldOffset has been called by
org.neo4j.driver.internal.shaded.io.netty.util.internal.PlatformDependent0$3
(file:/root/.m2/repository/org/neo4j/driver/neo4j-java-driver/4.4.5/neo4j-java-
driver-4.4.5.jar)
WARNING: Please consider reporting this to the maintainers of class
org.neo4j.driver.internal.shaded.io.netty.util.internal.PlatformDependent0$3
WARNING: sun.misc.Unsafe::objectFieldOffset will be removed in a future release
Waiting for Neo4j...
Waiting for Neo4j...
Waiting for Neo4j...
Waiting for Neo4j...
Waiting for Neo4j...
Waiting for Neo4j...
Waiting for Neo4j...
Waiting for Neo4j...
Neo4j is ready.
PROGRESS articles=1000 authors=2584 totalNodes=3584 elapsedSeconds=46
PROGRESS articles=2000 authors=5114 totalNodes=7114 elapsedSeconds=48
PROGRESS articles=3000 authors=7934 totalNodes=10934 elapsedSeconds=50
PROGRESS articles=4000 authors=11001 totalNodes=15001 elapsedSeconds=52
PROGRESS articles=5000 authors=14351 totalNodes=19351 elapsedSeconds=55
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PROGRESS articles=8000 authors=23265 totalNodes=31265 elapsedSeconds=60
PROGRESS articles=9000 authors=25713 totalNodes=34713 elapsedSeconds=61
PROGRESS articles=10000 authors=28309 totalNodes=38309 elapsedSeconds=62
PROGRESS articles=11000 authors=31230 totalNodes=42230 elapsedSeconds=64
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PROGRESS articles=14000 authors=40946 totalNodes=54946 elapsedSeconds=69
PROGRESS articles=15000 authors=43124 totalNodes=58124 elapsedSeconds=70
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PROGRESS articles=30000 authors=82299 totalNodes=112299 elapsedSeconds=93
PROGRESS articles=31000 authors=84716 totalNodes=115716 elapsedSeconds=94
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PROGRESS articles=33000 authors=90053 totalNodes=123053 elapsedSeconds=97
PROGRESS articles=34000 authors=92987 totalNodes=126987 elapsedSeconds=99
PROGRESS articles=35000 authors=95559 totalNodes=130559 elapsedSeconds=100
PROGRESS articles=36000 authors=97412 totalNodes=133412 elapsedSeconds=101
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PROGRESS articles=40000 authors=106871 totalNodes=146871 elapsedSeconds=106
PROGRESS articles=41000 authors=109749 totalNodes=150749 elapsedSeconds=109
PROGRESS articles=42000 authors=112291 totalNodes=154291 elapsedSeconds=110
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PROGRESS articles=51000 authors=132585 totalNodes=183585 elapsedSeconds=121
PROGRESS articles=52000 authors=134679 totalNodes=186679 elapsedSeconds=122
PROGRESS articles=53000 authors=136944 totalNodes=189944 elapsedSeconds=124
PROGRESS articles=54000 authors=139530 totalNodes=193530 elapsedSeconds=126
PROGRESS articles=55000 authors=142143 totalNodes=197143 elapsedSeconds=127
PROGRESS articles=56000 authors=144636 totalNodes=200636 elapsedSeconds=129
PROGRESS articles=57000 authors=146323 totalNodes=203323 elapsedSeconds=130
PROGRESS articles=58000 authors=148269 totalNodes=206269 elapsedSeconds=131
PROGRESS articles=59000 authors=150297 totalNodes=209297 elapsedSeconds=132
PROGRESS articles=60000 authors=152495 totalNodes=212495 elapsedSeconds=134
PROGRESS articles=61000 authors=154792 totalNodes=215792 elapsedSeconds=135
PROGRESS articles=62000 authors=157452 totalNodes=219452 elapsedSeconds=137
PROGRESS articles=63000 authors=159935 totalNodes=222935 elapsedSeconds=138
PROGRESS articles=64000 authors=161666 totalNodes=225666 elapsedSeconds=139
PROGRESS articles=65000 authors=163532 totalNodes=228532 elapsedSeconds=140
PROGRESS articles=66000 authors=165524 totalNodes=231524 elapsedSeconds=141
PROGRESS articles=67000 authors=167684 totalNodes=234684 elapsedSeconds=142
PROGRESS articles=68000 authors=170074 totalNodes=238074 elapsedSeconds=143
PROGRESS articles=69000 authors=172607 totalNodes=241607 elapsedSeconds=145
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PROGRESS articles=74000 authors=182589 totalNodes=256589 elapsedSeconds=151
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PROGRESS articles=104000 authors=243295 totalNodes=347295 elapsedSeconds=190
PROGRESS articles=105000 authors=245606 totalNodes=350606 elapsedSeconds=192
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PROGRESS articles=108000 authors=250493 totalNodes=358493 elapsedSeconds=195
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PROGRESS articles=110000 authors=254353 totalNodes=364353 elapsedSeconds=198
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PROGRESS articles=120000 authors=273464 totalNodes=393464 elapsedSeconds=211
PROGRESS articles=121000 authors=274901 totalNodes=395901 elapsedSeconds=212
PROGRESS articles=122000 authors=276394 totalNodes=398394 elapsedSeconds=213
PROGRESS articles=123000 authors=278156 totalNodes=401156 elapsedSeconds=213
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PROGRESS	articles=129000	authors=289008	totalNodes=418008	elapsedSeconds=221
PROGRESS	articles=130000	authors=290696	totalNodes=420696	elapsedSeconds=222
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PROGRESS	articles=132000	authors=294578	totalNodes=426578	elapsedSeconds=225
PROGRESS	articles=133000	authors=296767	totalNodes=429767	elapsedSeconds=227
PROGRESS	articles=134000	authors=298493	totalNodes=432493	elapsedSeconds=229
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PROGRESS	articles=137000	authors=302968	totalNodes=439968	elapsedSeconds=232
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PROGRESS	articles=985000	authors=1230597	totalNodes=2215597	elapsedSeconds=3704
PROGRESS	articles=986000	authors=1231537	totalNodes=2217537	elapsedSeconds=3723
PROGRESS	articles=987000	authors=1232463	totalNodes=2219463	elapsedSeconds=3734


```
PROGRESS articles=988000 authors=1233025 totalNodes=2221025 elapsedSeconds=3746
PROGRESS articles=989000 authors=1233676 totalNodes=2222676 elapsedSeconds=3757
PROGRESS articles=990000 authors=1234422 totalNodes=2224422 elapsedSeconds=3766
PROGRESS articles=991000 authors=1235166 totalNodes=2226166 elapsedSeconds=3782
PROGRESS articles=992000 authors=1236074 totalNodes=2228074 elapsedSeconds=3803
PROGRESS articles=993000 authors=1237104 totalNodes=2230104 elapsedSeconds=3819
PROGRESS articles=994000 authors=1237818 totalNodes=2231818 elapsedSeconds=3826
PROGRESS articles=995000 authors=1238466 totalNodes=2233466 elapsedSeconds=3838
PROGRESS articles=996000 authors=1239204 totalNodes=2235204 elapsedSeconds=3846
PROGRESS articles=997000 authors=1239968 totalNodes=2236968 elapsedSeconds=3855
PROGRESS articles=998000 authors=1240649 totalNodes=2238649 elapsedSeconds=3866
PROGRESS articles=999000 authors=1241496 totalNodes=2240496 elapsedSeconds=3892
PROGRESS articles=1000000 authors=1242546 totalNodes=2242546 elapsedSeconds=3910
LOAD_END=2026-05-03T21:59:26.386887899Z
ARTICLES_LOADED=1000000
AUTHORS_LOADED=1242546
TOTAL_NODES=2242546
DURATION_SECONDS=3910
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 01:05 h
[INFO] Finished at: 2026-05-03T21:59:27Z
[INFO] -----
PS C:\Users\LocalAdmin\MSE-
Local\TSM_AdvDaBa\TP02_Large_Database_Experiment_with_Neo4j\mse-advDaBa-Labo2>
```